

# End-to-End Delay

- ❑ **Transmission Delay**: The time taken for the bits of a packet to be placed on the wire
- ❑ **Processing Delay**: The time taken for inspecting the header bits to be matched with entries in forwarding table + detecting errors
- ❑ **Queuing Delay**: The time spent in the queue before it is processed
- ❑ **Propagation Delay**: The time taken for the bits to sail through the wire and reach other end of the wire

# Overall Delay

❑ Delay at a node

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

❑ Variable and fixed delay

❑ Delay at a node multiplied by the number of routers

$$d_{\text{end-end}} = N (d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$$

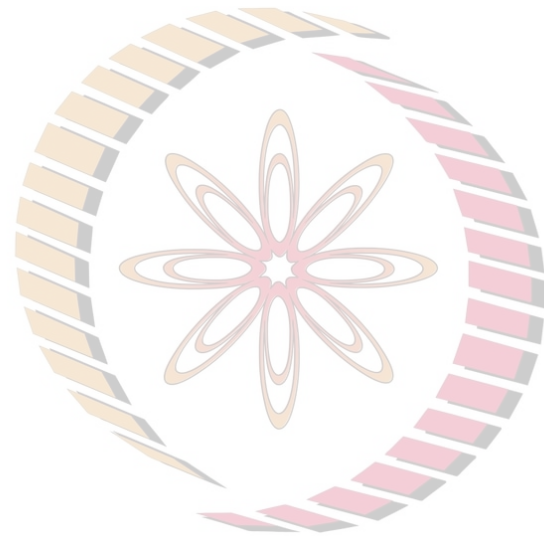
Assuming network is free of congestion

# Overall Delay

❑ Host A wants to send a 2 MB file to Host B over a single link. The link has the following characteristics:

- ❑ Transmission rate (R): 4 Mbps
- ❑ Distance between A and B: 2000 km
- ❑ Propagation speed:  $2 \times 10^8$  m/s
- ❑ Processing delay at each router: 2 ms
- ❑ There is one router between A and B
- ❑ Average queuing delay at the router: 5 ms
- ❑ The file is divided into 1000-byte packets

1. Calculate the transmission delay per packet.
2. Calculate the propagation delay per link.
3. Calculate the total processing delay for a packet.
4. Calculate the total queuing delay experienced.
5. What is the end-to-end delay for the first packet?
6. Approximately how long will it take to transfer the entire file (assume continuous transmission)?



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## 1) Transmission Delay per Packet

$$\text{Transmission delay} = \frac{\text{Packet size}}{\text{Rate}} = \frac{8000}{4 \times 10^6} = 2 \times 10^{-3} \text{ s} = 2 \text{ ms}$$

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## 2) Propagation Delay per Link

There are **2 links** (A → Router, Router → B), each assumed half the total distance:

$$\text{Distance per link} = \frac{2000}{2} = 1000 \text{ km} = 10^6 \text{ m}$$

$$\text{Propagation delay} = \frac{10^6}{2 \times 10^8} = 5 \times 10^{-3} \text{ s} = 5 \text{ ms}$$

### 3) Total Processing Delay

One router and hence one time processing

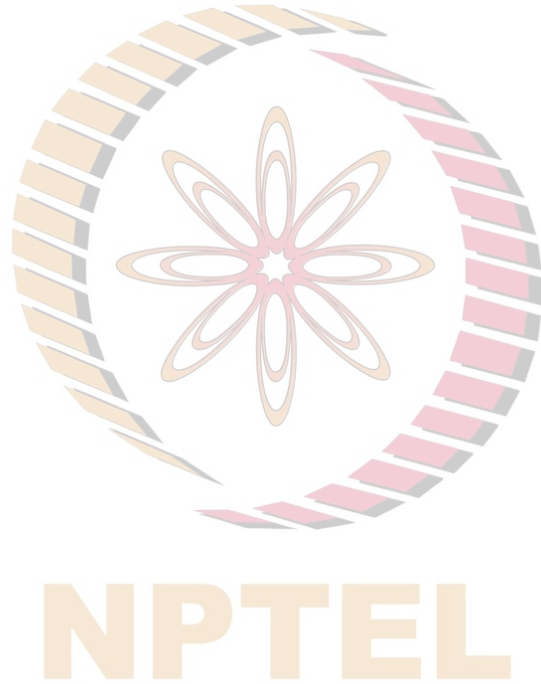
Processing delay=2 ms



## 4) Total Queuing Delay

One router and hence one queue

Queuing delay=5 ms



## 5) End-to-End Delay (First Packet)

$$\text{Total delay} = d_{\text{tran}} + d_{\text{prop}} + d_{\text{que}} + d_{\text{proc}}$$

$$d = 2 \times 2\text{ms} + 2 \times 5\text{ms} + 5\text{ms} + 2\text{ms} = 21\text{ms}$$

## 6) Total Time to Transfer Entire File

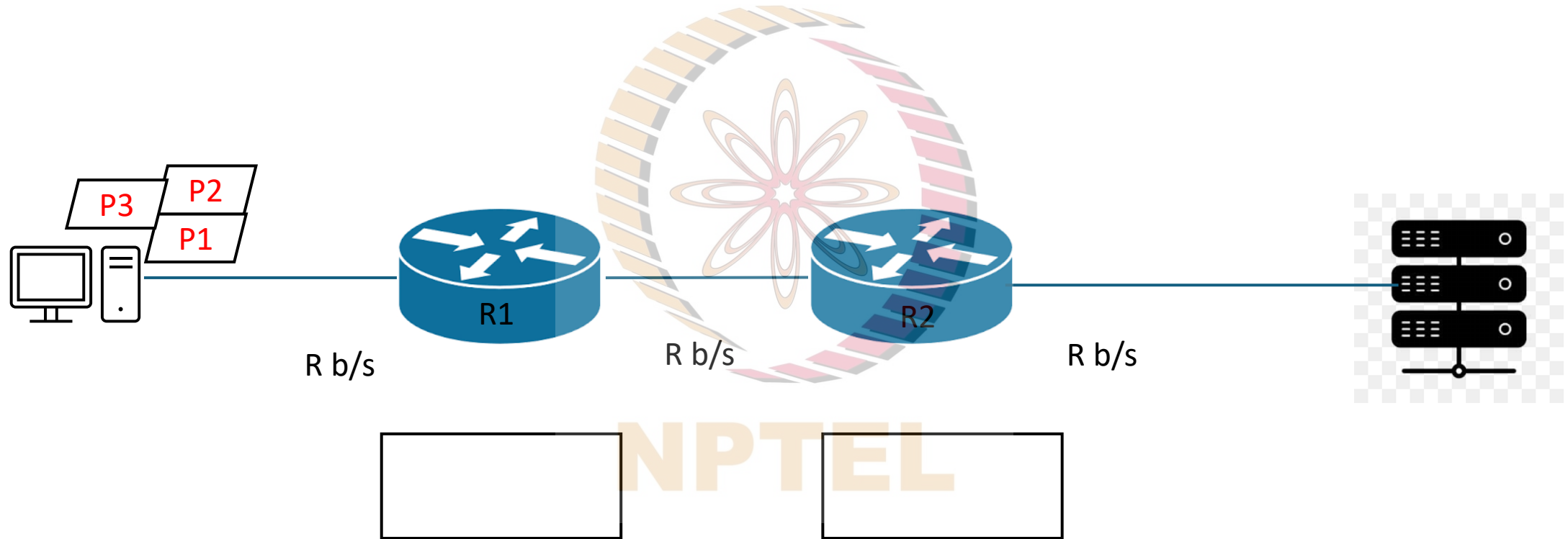
$$\text{Number of packets} = \frac{2 \times 10^6}{1000} = 2000 \text{ packets}$$

- ❑ Time to send all packets (pipeline effect)
- ❑ First packet takes 21 ms
- ❑ Remaining packets are sent every transmission delay = 2 ms

$$\begin{aligned} \text{Total time} &= 21 + (2000 - 1) \times 2 \text{ ms} \\ &= 21 + 1999 \times 2 = 21 + 3998 = 4019 \text{ ms} \end{aligned}$$

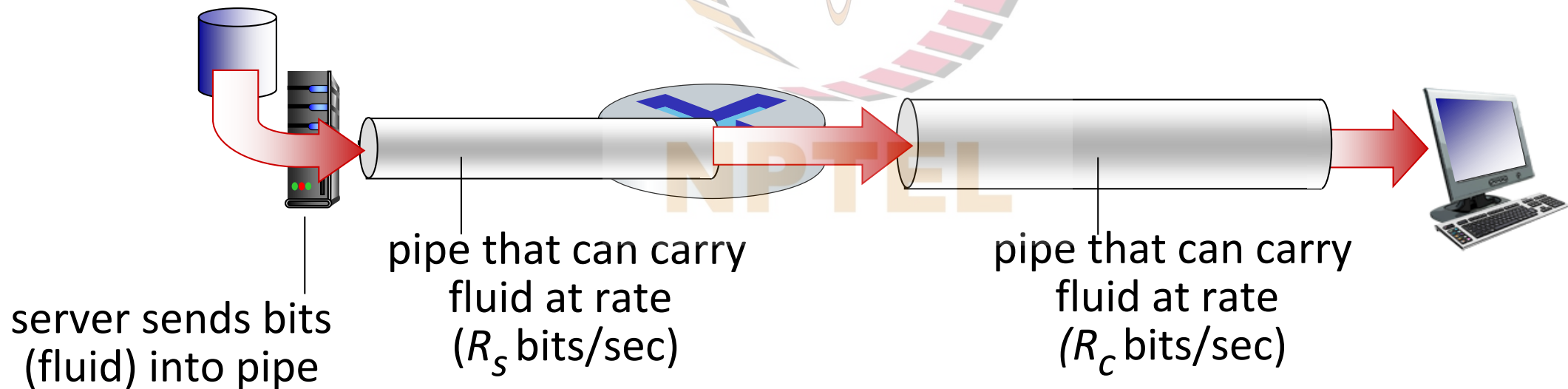
$$\approx 4.02 \text{ seconds}$$

# Transmission Delay: Multiple Packets and Multiple Hops



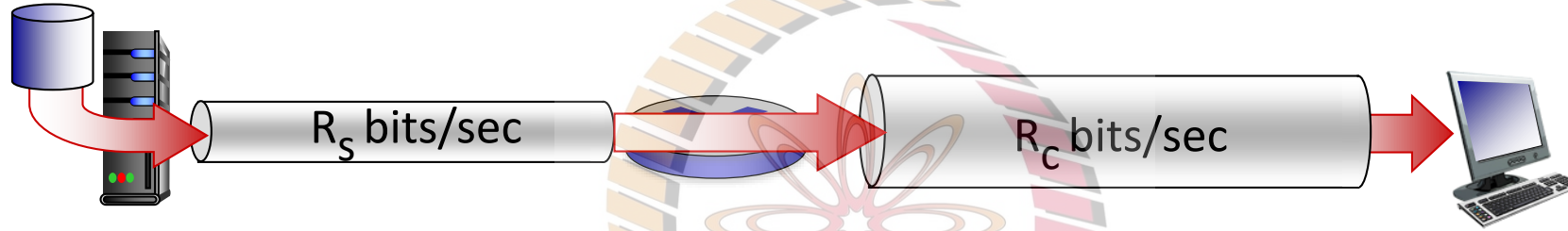
# Throughput

- *throughput*: rate (bits/time unit) at which bits are being sent from sender to receiver
  - *instantaneous*: rate at given point in time
  - *average*: rate over longer period of time

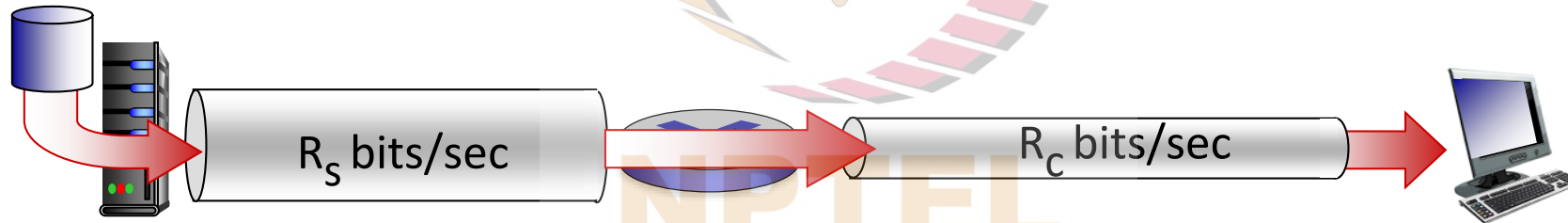


# Throughput

$R_s < R_c$  What is average end-end throughput?



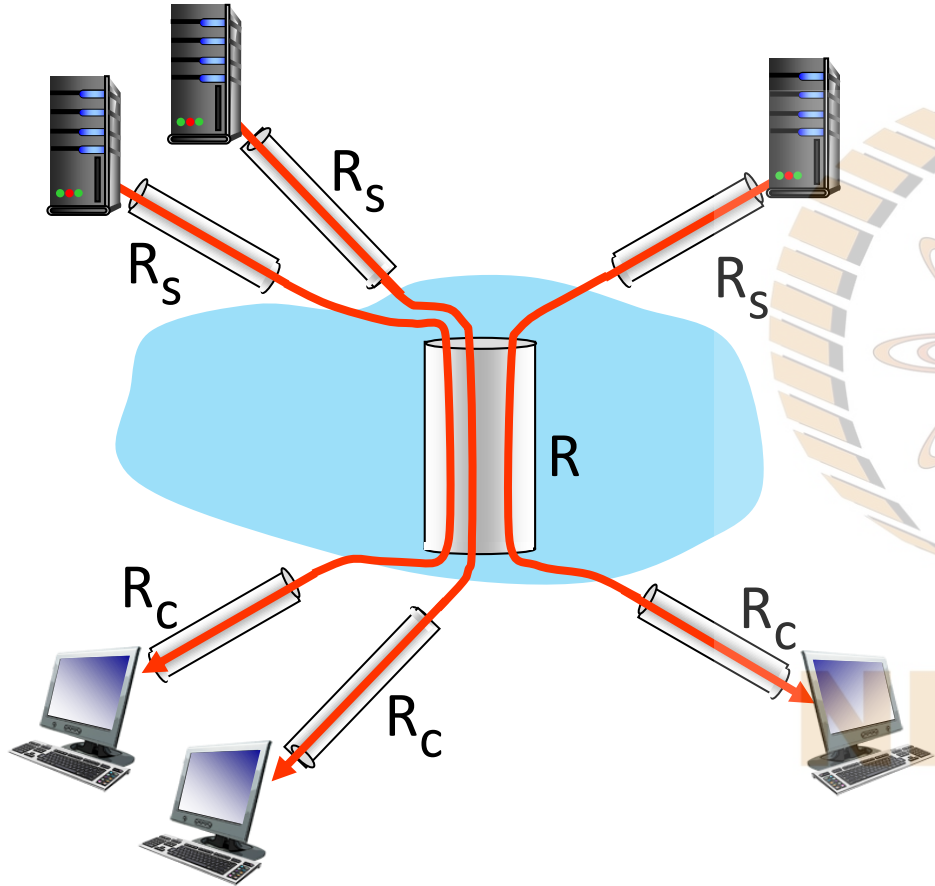
$R_s > R_c$  What is average end-end throughput?



*bottleneck link*

link on end-end path that constrains end-end throughput

# Throughput: network scenario



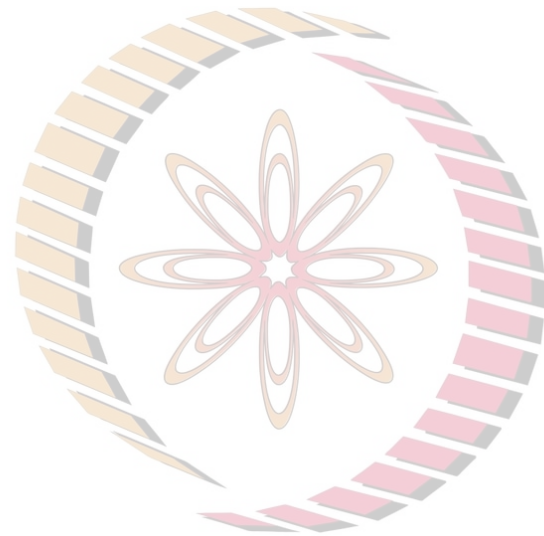
- per-connection end-end throughput:  
 $\min(R_c, R_s, R/10)$
- in practice:  $R_c$  or  $R_s$  is often bottleneck

10 connections (fairly) share  
backbone bottleneck link  $R$  bits/sec

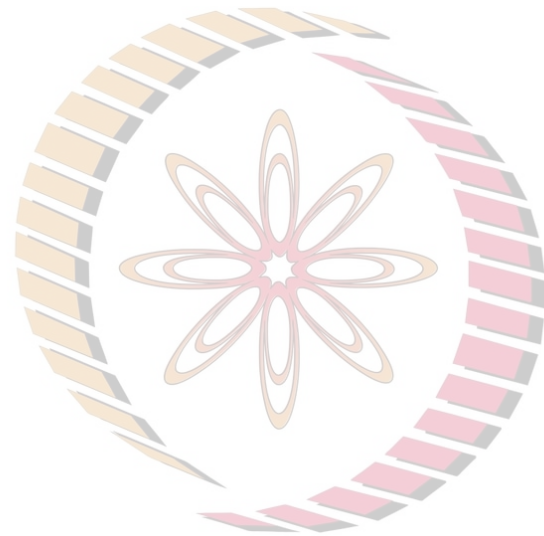
# Throughput

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates  $R1 = 500$  kbps,  $R2 = 2$  Mbps, and  $R3 = 1$  Mbps.

- a. Assuming no other traffic in the network, what is the throughput for the file transfer?
- b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?



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